

Conducted the first end-to-end system test for both Tomato Disease Detection and Crop Recommendation modules.

Successfully verified:

- Flutter mobile application interface
- Backend API communication using FastAPI/Flask
- ViT-Base disease detection inference
- LSTM crop recommendation inference
- PostgreSQL database integration
- Real-time result display in the mobile application
- Weather and environmental data retrieval
- User input handling and prediction workflow

System Performance:

- Disease detection response time under 5 seconds
- Crop recommendation response time under 5 seconds
- Stable operation on low-resource devices (2GB RAM)

All core functionalities operated successfully during integration testing with no critical system failures.